made in explaining the properties of light emitted by glowing-hot hydrogen atoms. But this and other work prior to the mid-1920s was more a makeshift union of nineteenth-century ideas with newfound quantum concepts than a coherent framework for understanding the physical universe. Compared with the clear, logical framework of Newton's laws of motion or Maxwell's electromagnetic theory, the partially developed quantum theory was in a chaotic state.

In 1923, the young French nobleman Prince Louis de Broglie added a new element to the quantum fray, one that would shortly help to usher in the mathematical framework of modern quantum mechanics and that earned him the 1929 Nobel Prize in physics. Inspired by a chain of reasoning rooted in Einstein's special relativity, de Broglie suggested that the wave-particle duality applied not only to light but to matter as well. He reasoned, roughly speaking, that Einstein's $E = mc$

2 relates mass to energy, that Planck and Einstein had related energy to the frequency of waves, and therefore, by combining the two, mass should have a wave-like incarnation as well. After carefully working through this line of thought, he suggested that just as light is a wave phenomenon that quantum theory shows to have an equally valid particle description, an electron—which we normally think of as being a particle—might have an equally valid description in terms of waves. Einstein immediately took to de Broglie's idea, as it was a natural outgrowth of his own contributions of relativity and of photons. Even so, nothing is a substitute for experimental proof. Such proof was soon to come from the work of Clinton Davisson and Lester Germer.

In the mid-1920s, Davisson and Germer, experimental physicists at the Bell telephone company, were studying how a beam of electrons bounces off of a chunk of nickel. The only detail that matters for us is that the nickel crystals in such an experiment act very much like the two slits in the experiment illustrated by the figures of the last section—in fact, it's perfectly OK to think of this experiment as being the same one illustrated there, except that a beam of electrons is used in place of a beam of light. We will adopt this point of view. When Davisson and Germer examined electrons making it through the two slits in the barrier by allowing them to hit a phosphorescent screen that recorded the location of impact of each electron by a bright dot—